

Michael S. Ildefonso

Full-Stack Developer | Backend Architecture | Secure System Design

Taytay, Rizal | 09457517383 | michaelildefonso20@gmail.com

linkedin.com/in/michael-ildefonso | github.com/michaelildefonso | michaelildefonso.ildf.site

PROFESSIONAL SUMMARY

Backend-focused Full-Stack Developer and final-year Computer Science student with hands-on experience in REST API design, database architecture, and secure authentication systems. Demonstrated ability to deliver independently — including a PyPI-published security library and a fine-tuned bilingual speech AI built on a self-curated 20-hour dataset. Proficient in Python, JavaScript, FastAPI, Vue 3, Node.js, and PostgreSQL.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Vue 3, Vite, FastAPI, Node.js, Express, Tailwind CSS, Material UI, SQLAlchemy

Databases: PostgreSQL

Auth & Security: OAuth 2.0, JWT, Argon2id, Rate Limiting, Password Hashing, CORS, SSL

AI & Machine Learning: Model Fine-Tuning, faster-whisper, NLLB-200, Qwen2.5, FFmpeg, HuggingFace

DevOps & Cloud: Git, GitHub, GitHub Actions, CI/CD, Vercel, Railway, Render, Cloudflare R2, Modal

PROJECTS

Rekapo — Bilingual AI Summarizer

2025 – 2026 | Open Source | Capstone Thesis

Python, FastAPI, faster-whisper, NLLB-200 1.3B, Qwen2.5-1.5B, React Native, PostgreSQL, Google OAuth 2.0, GitHub Actions, FFmpeg, Railway, Modal

Repos: [API](#) | [Mobile App](#) | [Admin](#)

- Architected and developed the full backend and system design (ERD, DFDs, use case modeling) for a bilingual AI application to transcribe and summarize Taglish speech, a language underserved by standard STT models
- Curated a 20-hour bilingual dataset from scratch and fine-tuned Whisper-small to outperform the Whisper-Medium baseline; applied float16 quantization to meet deployment constraints

Progresstify — Kanban Project Management Tool

2025 | Open Source | Academic Team Project

Node.js, Express, PostgreSQL, Neon.tech, OAuth 2.0, JWT, Vite, Vercel, Render

Repos: [Frontend](#) | [Backend](#)

- Implemented multi-provider OAuth 2.0 (Google and GitHub) with JWT from scratch; used serverless PostgreSQL on Neon.tech for a stateless, independently scalable data layer
- Separated frontend and API deployments across Vercel and Render; built a standalone admin panel for user and system-wide management

bisHash — Argon2id Password Hashing Library

2024 | Open Source | Published on PyPI

Python, Argon2id, PyPI, Rate Limiting, Security Engineering

Repos: [Library](#) | [Demo](#)

- Engineered and published a Python security library on PyPI implementing Argon2id with server-side peppering, IP-based rate limiting, and brute-force lockout — addressing a gap in standard bcrypt-based authentication
- Abstracted Argon2id configuration, peppering, and rate limiting behind a minimal two-function API — reducing the implementation surface for secure password storage

EDUCATION

ACL College — Taytay, Rizal, Philippines

Expected Graduation: May 2026

Summa Cum Laude

Bachelor of Science in Computer Science | GPA: 1.16 (Scale: 1.0 = Highest)